

QXU6033 Advanced Polymer Chemistry

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1. Introduction to Module

■ Assessment

Activity	Score
Professional Practice Evaluation(supervision)	20%
Coursework	30%
Final Exam	50%

1.1. Study Goal

■ Controlled Radical Polymerisation

Mechanistic principles of controlled / living radical polymerisation (CRP)
 Kinetic control, equilibrium processes, and radical deactivation strategies
 Molecular weight control, dispersity ($\bar{D}$), and end-group fidelity

■ Core CRP methodologies

Nitroxide-Mediated Polymerisation (NMP)
 Reversible Addition–Fragmentation Chain Transfer (RAFT)
 Atom Transfer Radical Polymerisation (ATRP)

■ Advanced macromolecular architectures

Self-Condensing Vinyl Polymerisation (SCVP)
 Strathclyde synthesis and hyperbranched structures
 Star polymers and multi-arm architectures

■ From synthesis to structure

Block copolymer design and synthesis
 Microphase separation and self-assembly

Introductory Discussions I

Q: A has a structure that enables it to polymerise through chain-growth polymerisation, while B is step-growth.

Scenario 1: Imagine you start polymerising A, and half-way through the polymerisation, you add C, also another monomer, which can polymerise through chain-growth polymerisation, and, that can polymerise together well with A (this is called “copolymerise”). Separately, imagine Scenario 2: Use B instead A, then add D, a monomer, which can polymerise through step-growth polymerisation, and, that can copolymerise well with B. At the end of both polymerisation processes, what type of polymer samples will you have in each Scenario?

A: S1: AAAAAA-(random A and C)-(if one is finish, pure another one).

S2:(block B)-some D-(block B) and cycle B.

2. Controlled Radical Polymerisation

Note

CRP, controlled / living radical polymerisation.

Also Reversible Deactivation Radical Polymerisation, RDRP.

2.1. Towards Living Polymerisation

As long as “living” polymer is kept tightly closed (no H₂O, ROH, CO₂, O₂) more monomer can be added and the molecular weight will increase.

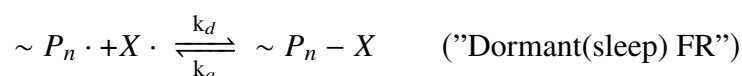
1. MW increases steadily with time/conversion
2. Narrow PDI (< 1.2 or 1.3)
3. Can add more monomer or another monomer/blocks.

■ Kinetic Issue

To make FRP “living”, need to inhibit termination and chain-transfer without affecting initiation/propagation too much.

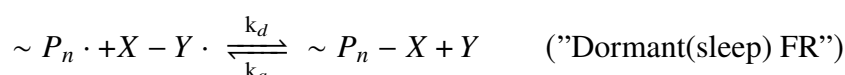
■ Strategy 1: Reversible Termination

Also, Reversible Deactivation.



Note: Do not use Strategy 1, either refer to specific chemistry (NMP, or TEMPO) or “reversible deactivation”

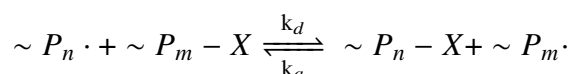
■ Strategy 2: Catalytic/Assisted Reversible Termination



X can be single atom transferred to P·, Y normally a transition metal.

Also, **Catalytic Reversible Deactivation.**

■ Reversible Chain-Transfer(Degenerative Transfer)



In processes $\sim P_n \cdot$ or $\sim P_m \cdot$ themselves, **Termination still occurs.**

To be effective

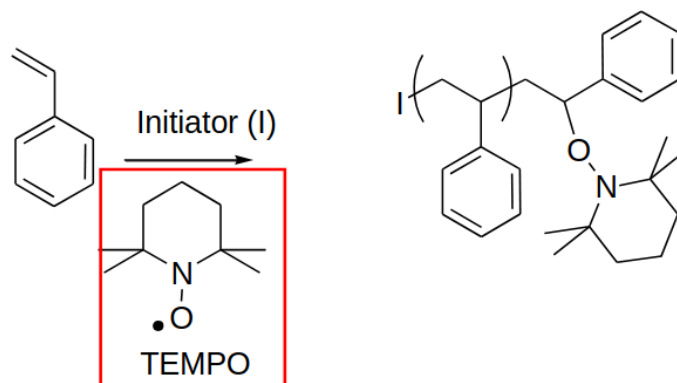
Reversible Termination should be $k_d > k_a$

Reversible Chain-Transfer is $k_d \sim k_a$

2.2. Controlled Radical Polymerisation by Reversible Deactivation

Specific chemistries:

Nitroxyl radicals (e.g. TEMPO) for Nitroxyl Mediated Polymerisation (NMP)



TEMPO: (2,2,6,6-Tetramethylpiperidin-1-yl)oxyl radical

Note

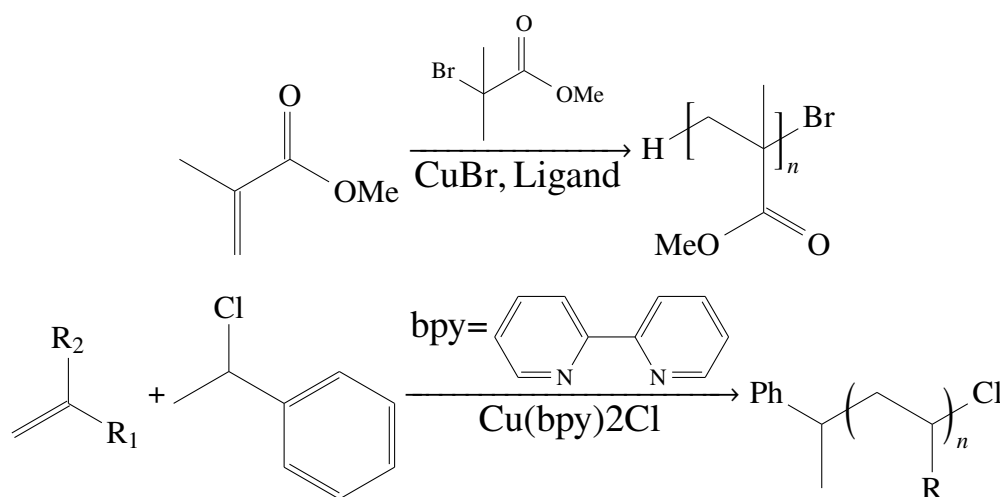
The equilibrium with TEMPO doesn't happen necessarily at every single monomer addition.

2.3. Controlled Radical Polymerisation by Catalytic Reversible Deactivation

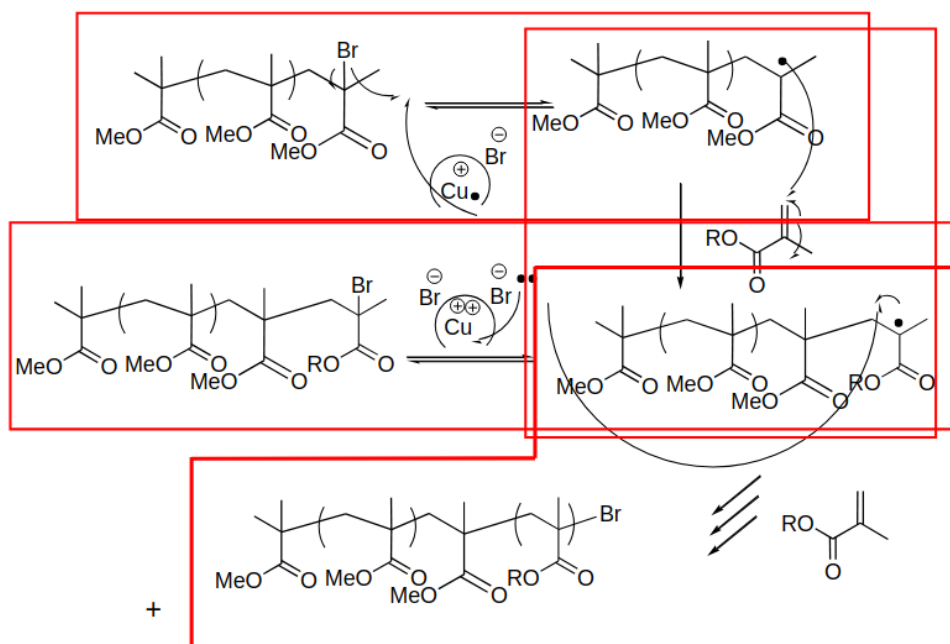
Specific chemistries:

■ ATRP (Atom-transfer free radical polymerisation)

A bromo (or chloro) compound + Cu^+ complex act as the initiator: does not require an external free-radical initiation system.



End-groups depend on initiator.



Advantages:

- Simple
- No need to heat up
- Tolerant of a range of functional groups

Disadvantages:

- Difficult to replicate literature results
- Very sensitive to O₂ (will oxidise Cu⁺ → Cu⁺⁺, stopping activation)
- Amino/amide/COOH-containing monomers compete for ligand and change kinetics (usually not favourably)
- Cu is toxic (removal of Cu from biomaterials is difficult)

■ ARGET-ATRP(Activators Re-Generated by Electron Transfer)

Using external reducing agents to convert Cu⁺⁺ → Cu⁺. So, more tolerant of O₂ and low quantity of Cu.

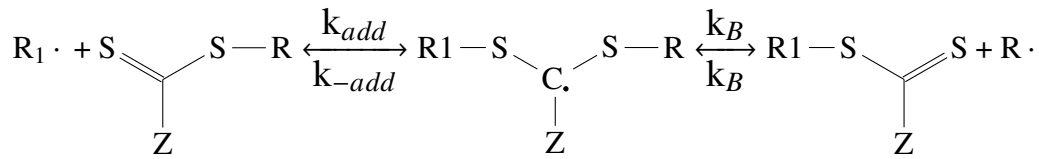
■ SARA-ATRP(Supplement Activation Reducing Agent)

Addition of Cu(I) to both reduce Cu⁺⁺ → Cu⁺, and active C-Br/C-Cl to C• (FR).

2.4. Reversible Addition-Fragmentation Chain-Transfer Polymerisation

Requires an external free radical initiator and a RAFT agent (dithioester, or similar species)

■ How Agents Work



3. Non-Vinyl Hyperbranched Polymer

3.1. Branching in Step-Growth

■ Carothers for linear system

$$\overline{DP} = \frac{1+r}{1+r-2rp}$$

■ Carothers for branched system

$$\overline{DP} = \frac{2}{2 - pf_{av}}$$

f_{av} : Functional factor, average number of functional groups per monomer, in linear chain: = 2.

$$f_{av} = \frac{N_A f_A + N_B f_B + \cdots + N_i f_i}{N_A + N_B + \cdots + N_i}$$

N_A : number of moles of monomer A,

f_A : functionality of monomer A.

In Ring-Opening: only branching, no crosslinking, so added functional group has only 1 functionality.

3.2. Gel Point

At **Gel Point**, the system becomes a gel: a crosslinked polymer.

gels are soft solids swollen with a liquid, in this particular case, just crosslinked and also presence of solvent or liquid monomer softens the material. Higher crosslinking and removal of solvent hardens the material.

$$p_G = \frac{2}{f_{av}}$$

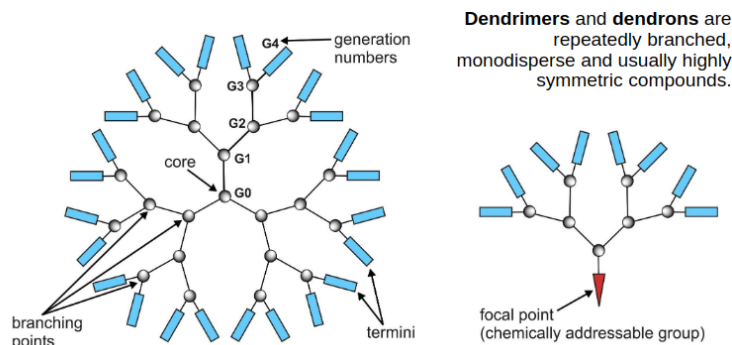
p_G : conversion at gel point.

Note

Carothers theory overestimates gel point (experimental < calculated p_G).

4. Dendrimers

Dendrimers are repeatedly branched molecules.



5. Hyperbranch through Self-Condensing Vinyl Polymerisation

■ Hyperbranched Polymers vs Dendrimers

Hyperbranched Polymers:

- ✓ “One-pot” reactions (reactions that lead to desired product in just one reaction)
- ✓ Intermediate properties (viscosity)
- × Large polydispersity
- × Not so good control over end-groups and branching

Dendrimers:

- ✓ Monodisperse polymers with defined branching and end-groups
- ✓ Excellent/extreme properties (viscosity)
- ✓ Excellent carriers (cavities or pockets)
- × Very difficult/costly synthesis over a large series of reactions

5.1. Characterisation of Branching

■ Mark-Houwink-Sakurada Equation

$$[\eta] = KM^\alpha$$

Where,

1. $[\eta]$: Intrinsic viscosity.
2. M : Molecular weight of the polymer
3. K, a : Constants that depend on solvent, temperature and type of polymer.

6. Hyperbranched Polymers Using Cross-Linkers

7. Star Polymers

Many polymer "arms" from a central "core" forming a star shaped polymer, called **Star Polymers**.

■ Miktoarm

"Mixed" arms, which means different types of arms onto a core.

7.1. Synthesis Methods of Star Polymers

■ Core First

Begin from central multi-functional initiator.

Using RDRP(or CRP):

1. No termination
2. No chain transfer
3. Fast initiation and controlled propagation(to make sure all arms have similar length).
4. Enable block copolymer arms.

However:

1. Low number of arms per star molecules(3-8)
2. No Miktoarm
3. Can no design arms separately.

■ Arm First

Synthesis polymer arm by CRP,